Homework 1: Potential Fields and Boundary Value Problems

This homework covers material from **Module 1: Potential Fields, Laplace's Equation, and Boundary Value Problems** and aligns with learning outcomes **LO1**.

Related Resources: - Lectures: M1/lecture1_notes.qmd (Potential Fields and Laplace's Equation), M1/lecture2_notes.qmd (Green's Identities) - Lab: Lab 1 (Solving Laplace's Equation Numerically) - Textbook: Blakely Ch. 1–2

Problems

1. **Problem 1.1** (Blakely Ch. 1, Problem 1) — *Foundation/Moderate*

   - Show that for a conservative field, the work done around a closed path is zero. Derive this from the definition of a potential.
   - **Cross-reference:** M1/lecture1_notes.qmd, Section "What is a Potential Field?"
   - **Difficulty:** Moderate (requires calculus and path integration)
   - **Tip:** Start with line integral definition of work; use the gradient theorem

2. **Problem 1.2** (Blakely Ch. 2, Problem 3) — *Conceptual/Moderate*

   - Verify Green's second identity for two functions u = r and v = 1/r in a spherical volume. What does this tell us about the behavior of harmonic functions?
   - **Cross-reference:** M1/lecture2_notes.qmd, Section "Green's Identities"
   - **Difficulty:** Moderate (requires vector calculus and spherical coordinates)
   - **Tip:** Recall that 1/r is harmonic (^{2}(1/r) = 0) and r is not

3. **Problem 1.3** (Blakely Ch. 2, Problem 5) — *Classification/Moderate*

   - Classify the following boundary value problems and determine if they have unique solutions:
     - Dirichlet problem: = f on boundary, ^{2} = 0 inside
     - Neumann problem: / n = g on boundary, ^{2} = 0 inside
     - Robin problem: / n + h = g on boundary, ^{2} = 0 inside
   - **Cross-reference:** M1/lecture2_notes.qmd, Section "Green's Identities and Boundary Value Problems"
   - **Difficulty:** Moderate (classification and uniqueness analysis)
   - **Tip:** Use Green's identities to prove uniqueness; note that Neumann has solution unique up to constant

4. **Problem 1.4** (Blakely Ch. 1, Problem 2) — *Proof/Moderate*

   - Prove that harmonic functions satisfy the maximum principle. What are the implications for potential field interpretation?

- **Cross-reference:** M1/lecture1_notes.qmd, Section "Harmonic Functions"; Lab 1 (numerical verification)
- **Difficulty:** Moderate (proof technique; conceptual application)
- **Tip:** Start with Gauss's mean value theorem; apply to contradiction argument

5. **Problem 1.5** — *Conceptual/Challenging*

   - Explain why potential field data interpretation is inherently nonunique. Use Green's identities to illustrate your point.
   - **Cross-reference:** M1/lecture2_notes.qmd, Section "Geophysical Implications"; Final Project (application)
   - **Difficulty:** Challenging (requires synthesis and conceptual depth)
   - **Tip:** Consider multiple source distributions producing same boundary data; relate to Green's theorem volume-to-surface transformation

Submission Guidelines

- Submit solutions as a PDF or Quarto document
- Include derivations, calculations, and explanations
- Use LaTeX for mathematical expressions
- Feel free to include sketches or diagrams
- Cite references (e.g., Blakely, lecture notes) where appropriate
- Expected length: 3–5 pages